

Unzip the data

```
In [1]: !unzip -q traffic_signs.zip
```

Installing and importing libraries

```
In [2]: !pip install ultralytics
```

Collecting ultralytics

Downloading ultralytics-8.3.248-py3-none-any.whl.metadata (37 kB)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.12.0.88)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.9.0+cu126)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.24.0+cu126)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.31.0)

Collecting ultralytics-thop>=2.0.18 (from ultralytics)

Downloading ultralytics_thop-2.0.18-py3-none-any.whl.metadata (14 kB)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (25.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2025.11.12)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.20.0)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.15.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.1)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025.3.0)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.80)

Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (9.10.2.21)

Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.3.0.4)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (10.3.7.77)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.7.1.2)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.5.4.2)

Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)

Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2.27.5)

Requirement already satisfied: nvidia-nvshmem-cu12==3.3.20 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.3.20)

Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.85)

Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.11.1.6)

Requirement already satisfied: triton==3.5.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.5.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)

Downloading ultralytics-8.3.248-py3-none-any.whl (1.2 MB)

1.2/1.2 MB 75.5 MB/s eta 0:00:00

Downloading ultralytics_thop-2.0.18-py3-none-any.whl (28 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.3.248 ultralytics-thop-2.0.18

```
In [3]: import os
import random
import pandas as pd
from PIL import Image
import cv2
from ultralytics import YOLO
```

```

from IPython.display import Video
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style='darkgrid')
import pathlib
import glob
from tqdm.notebook import trange, tqdm
import warnings
warnings.filterwarnings('ignore')

```

Creating new Ultralytics Settings v0.0.6 file 

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

In [4]: `sns.set(rc={'axes.facecolor': '#eae8fa'}, style='darkgrid')`

Exploratory Data Analysis (EDA)

```

In [5]: Image_dir = 'car/train/images'

num_samples = 9
image_files = os.listdir(Image_dir)

# Randomly select num_samples images
rand_images = random.sample(image_files, num_samples)

fig, axes = plt.subplots(3, 3, figsize=(11, 11))

for i in range(num_samples):
    image = rand_images[i]
    ax = axes[i // 3, i % 3]
    ax.imshow(plt.imread(os.path.join(Image_dir, image)))
    ax.set_title(f'Image {i+1}')
    ax.axis('off')

plt.tight_layout()
plt.show()

```

Image 1



Image 2



Image 3



Image 4



Image 5



Image 6

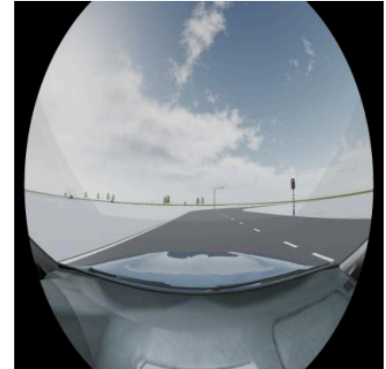


Image 7



Image 8



Image 9



```
In [6]: # Get the size of the image
image = cv2.imread("car/train/images/00000_00000_00012_png.rf.23f94508dba03ef2f8bd1
h, w, c = image.shape
print(f"The image has dimensions {w}x{h} and {c} channels.")
```

The image has dimensions 416x416 and 3 channels.

Try pre-trained YOLOv8

```
In [7]: # Use a pretrained YOLOv8n model
model = YOLO("yolov8n.pt")

# Use the model to detect object
image = "car/train/images/FisheyeCamera_1_00228_png.rf.e7c43ee9b922f7b2327b8a00ccf4
result_predict = model.predict(source = image, imgsz=(640))

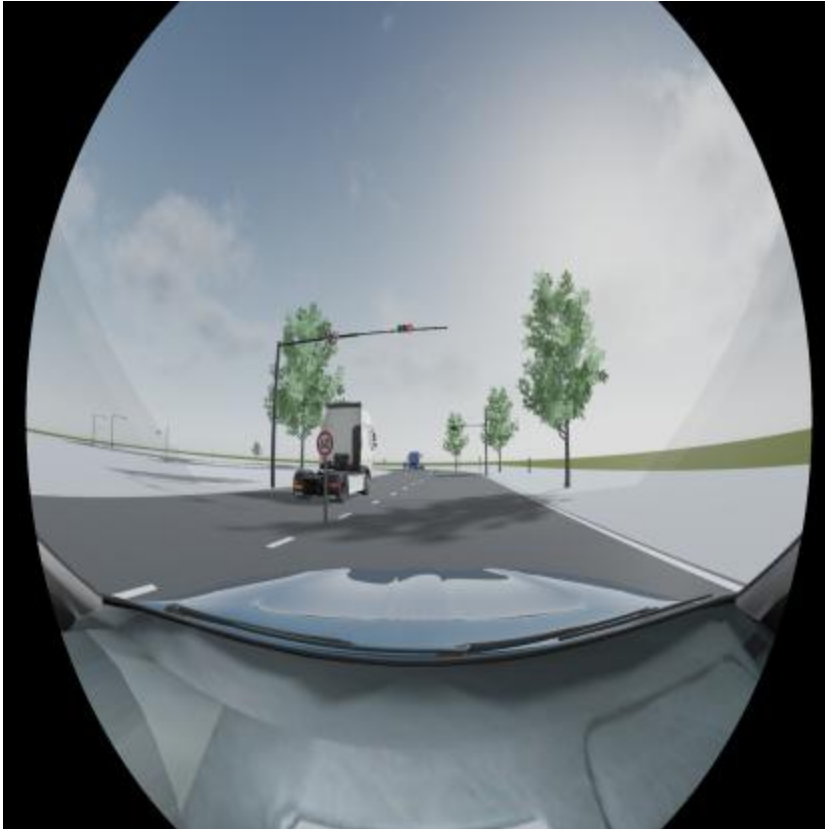
# show results
```

```
plot = result_predict[0].plot()  
plot = cv2.cvtColor(plot, cv2.COLOR_BGR2RGB)  
display(Image.fromarray(plot))
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.3.0/yolov8n.pt> to 'yolov8n.pt': 100% — 6.2MB 327.6MB/s 0.0s

image 1/1 /content/car/train/images/FisheyeCamera_1_00228_png.rf.e7c43ee9b922f7b2327b8a00ccf46a4c.jpg: 640x640 (no detections), 7.9ms

Speed: 13.8ms preprocess, 7.9ms inference, 4.6ms postprocess per image at shape (1, 3, 640, 640)



YOLOv8 based traffic signs detection model

- Training step

In [8]: `!pip install --upgrade ultralytics ray`

Requirement already satisfied: ultralytics in /usr/local/lib/python3.12/dist-packages (8.3.248)

Collecting ray

Downloading ray-2.53.0-cp312-cp312-manylinux2014_x86_64.whl.metadata (22 kB)

Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.2)

Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (3.10.0)

Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (4.12.0.88)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (11.3.0)

Requirement already satisfied: pyyaml>=5.3.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (6.0.3)

Requirement already satisfied: requests>=2.23.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.32.4)

Requirement already satisfied: scipy>=1.4.1 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.16.3)

Requirement already satisfied: torch>=1.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.9.0+cu126)

Requirement already satisfied: torchvision>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (0.24.0+cu126)

Requirement already satisfied: psutil>=5.8.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (5.9.5)

Requirement already satisfied: polars>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (1.31.0)

Requirement already satisfied: ultralytics-thop>=2.0.18 in /usr/local/lib/python3.12/dist-packages (from ultralytics) (2.0.18)

Requirement already satisfied: click>=7.0 in /usr/local/lib/python3.12/dist-packages (from ray) (8.3.1)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from ray) (3.20.0)

Requirement already satisfied: jsonschema in /usr/local/lib/python3.12/dist-packages (from ray) (4.25.1)

Requirement already satisfied: msgpack<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from ray) (1.1.2)

Requirement already satisfied: packaging>=24.2 in /usr/local/lib/python3.12/dist-packages (from ray) (25.0)

Requirement already satisfied: protobuf>=3.20.3 in /usr/local/lib/python3.12/dist-packages (from ray) (5.29.5)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (4.61.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (1.4.9)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.3.0->ultralytics) (2.9.0.post0)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.4.4)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.23.0->ultralytics) (2025.11.12)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (4.15.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (75.2.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.14.0)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.6.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.1.6)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2025.3.0)

Requirement already satisfied: NVIDIA-CUDA-nvrtc-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: NVIDIA-CUDA-runtime-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: NVIDIA-CUDA-cupti-cu12==12.6.80 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.80)

Requirement already satisfied: NVIDIA-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (9.10.2.21)

Requirement already satisfied: NVIDIA-cublas-cu12==12.6.4.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.4.1)

Requirement already satisfied: NVIDIA-cufft-cu12==11.3.0.4 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.3.0.4)

Requirement already satisfied: NVIDIA-curand-cu12==10.3.7.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (10.3.7.77)

Requirement already satisfied: NVIDIA-cusolver-cu12==11.7.1.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (11.7.1.2)

Requirement already satisfied: NVIDIA-cusparse-cu12==12.5.4.2 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.5.4.2)

Requirement already satisfied: NVIDIA-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (0.7.1)

Requirement already satisfied: NVIDIA-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (2.27.5)

Requirement already satisfied: NVIDIA-nvshmem-cu12==3.3.20 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.3.20)

Requirement already satisfied: NVIDIA-nvtx-cu12==12.6.77 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.77)

Requirement already satisfied: NVIDIA-nvjitlink-cu12==12.6.85 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (12.6.85)

Requirement already satisfied: NVIDIA-cufile-cu12==1.11.1.6 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (1.11.1.6)

Requirement already satisfied: triton==3.5.0 in /usr/local/lib/python3.12/dist-packages (from torch>=1.8.0->ultralytics) (3.5.0)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema->ray) (25.4.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema->ray) (2025.9.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema->ray) (0.37.0)

Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from jsonschema->ray) (0.30.0)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.3.0->ultralytics) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.8.0->ultralytics) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.8.0->ultralytics) (3.0.3)
Downloading ray-2.53.0-cp312-cp312-manylinux2014_x86_64.whl (72.4 MB)
_____ 72.4/72.4 MB 13.6 MB/s eta 0:00:00

Installing collected packages: ray
Successfully installed ray-2.53.0

```
In [9]: # Build from YAML and transfer weights
        Final_model = YOLO('yolov8n.pt')

        # Training The Final Model
        Result_Final_model = Final_model.train(data="car/data.yaml", epochs = 30, batch = -1
```

Ultralytics 8.3.248 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
engine/trainer: agnostic_nms=False, amp=True, augment=False, auto_augment=randaugmen
t, batch=-1, bgr=0.0, box=7.5, cache=False, cfg=None, classes=None, close_mosaic=10,
cls=0.5, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=Fals
e, cutmix=0.0, data=car/data.yaml, degrees=0.0, deterministic=True, device=None, dfl
=1.5, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=30, erasing=0.4, exi
st_ok=False, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None,
half=False, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, int8=False, iou=0.7, keras
=False, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mi
xup=0.0, mode=train, model=yolov8n.pt, momentum=0.937, mosaic=1.0, multi_scale=Fals
e, name=train, nbs=64, nms=False, opset=None, optimize=False, optimizer=auto, overla
p_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True,
profile=False, project=None, rect=False, resume=False, retina_masks=False, save=Tru
e, save_conf=False, save_crop=False, save_dir=/content/runs/detect/train, save_frame
s=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=
0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, s
ingle_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=Non
e, tracker=botsort.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visual
ize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=
0.0005, workers=8, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/A
rial.ttf': 100% ————— 755.1KB 52.8MB/s 0.0s
Overriding model.yaml nc=80 with nc=15

	from	n	params	module	a
rguments					
0	-1	1	464	ultralytics.nn.modules.conv.Conv	
[3, 16, 3, 2]					
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	
[16, 32, 3, 2]					
2	-1	1	7360	ultralytics.nn.modules.block.C2f	
[32, 32, 1, True]					
3	-1	1	18560	ultralytics.nn.modules.conv.Conv	
[32, 64, 3, 2]					
4	-1	2	49664	ultralytics.nn.modules.block.C2f	
[64, 64, 2, True]					
5	-1	1	73984	ultralytics.nn.modules.conv.Conv	
[64, 128, 3, 2]					
6	-1	2	197632	ultralytics.nn.modules.block.C2f	
[128, 128, 2, True]					
7	-1	1	295424	ultralytics.nn.modules.conv.Conv	
[128, 256, 3, 2]					
8	-1	1	460288	ultralytics.nn.modules.block.C2f	
[256, 256, 1, True]					
9	-1	1	164608	ultralytics.nn.modules.block.SPPF	
[256, 256, 5]					
10	-1	1	0	torch.nn.modules.upsampling.Upsample	
[None, 2, 'nearest']					
11	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	
[1]					
12	-1	1	148224	ultralytics.nn.modules.block.C2f	
[384, 128, 1]					
13	-1	1	0	torch.nn.modules.upsampling.Upsample	
[None, 2, 'nearest']					
14	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	
[1]					

```

15          -1  1      37248  ultralytics.nn.modules.block.C2f
[192, 64, 1]
16          -1  1      36992  ultralytics.nn.modules.conv.Conv
[64, 64, 3, 2]
17          [-1, 12]  1          0  ultralytics.nn.modules.conv.Concat
[1]
18          -1  1     123648  ultralytics.nn.modules.block.C2f
[192, 128, 1]
19          -1  1     147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
20          [-1, 9]  1          0  ultralytics.nn.modules.conv.Concat
[1]
21          -1  1     493056  ultralytics.nn.modules.block.C2f
[384, 256, 1]
22          [15, 18, 21]  1     754237  ultralytics.nn.modules.head.Detect
[15, [64, 128, 256]]
Model summary: 129 layers, 3,013,773 parameters, 3,013,757 gradients, 8.2 GFLOPs

```

Transferred 319/355 items from pretrained weights

Freezing layer 'model.22.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

Downloading <https://github.com/ultralytics/assets/releases/download/v8.3.0/yolo11n.pt> to 'yolo11n.pt': 100% ————— 5.4MB 65.8MB/s 0.1s

AMP: checks passed 

train: Fast image access  (ping: 0.0±0.0 ms, read: 1038.6±384.2 MB/s, size: 24.1 KB)

train: Scanning /content/car/train/labels... 3530 images, 3 backgrounds, 0 corrupt: 100% ————— 3530/3530 2.3Kit/s 1.5s

train: New cache created: /content/car/train/labels.cache

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

AutoBatch: Computing optimal batch size for imgsz=640 at 60.0% CUDA memory utilization.

AutoBatch: CUDA:0 (Tesla T4) 14.74G total, 0.11G reserved, 0.08G allocated, 14.55G free

	Params	GFLOPs	GPU_mem (GB)	forward (ms)	backward (ms)	
input						
		output				
3013773	8.209	0.405	74.4	571.4	(1, 3, 64	
0, 640)		list				
3013773	16.42	0.545	26.63	264.2	(2, 3, 64	
0, 640)		list				
3013773	32.84	0.843	20.17	157.1	(4, 3, 64	
0, 640)		list				
3013773	65.67	1.437	31.87	167.5	(8, 3, 64	
0, 640)		list				
3013773	131.3	2.487	60.42	166.5	(16, 3, 64	
0, 640)		list				

AutoBatch: Using batch-size 62 for CUDA:0 9.10G/14.74G (62%) 

train: Fast image access  (ping: 0.0±0.0 ms, read: 1276.4±609.5 MB/s, size: 26.0 KB)

train: Scanning /content/car/train/labels.cache... 3530 images, 3 backgrounds, 0 corrupt: 100% ————— 3530/3530 6.3Mit/s 0.0s

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

val: Fast image access  (ping: 0.0±0.0 ms, read: 378.7±194.4 MB/s, size: 19.9 KB)
val: Scanning /content/car/valid/labels... 801 images, 0 backgrounds, 0 corrupt: 10
0% _____ 801/801 1.9Kit/s 0.4s
val: New cache created: /content/car/valid/labels.cache
Plotting labels to /content/runs/detect/train/labels.jpg...
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and dete
rmining best 'optimizer', 'lr0' and 'momentum' automatically...
optimizer: AdamW(lr=0.000526, momentum=0.9) with parameter groups 57 weight(decay=0.
0), 64 weight(decay=0.000484375), 63 bias(decay=0.0)
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/runs/detect/train
Starting training for 30 epochs...

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
1/30	7.5G	0.8818	3.738	1.212	133	640: 100%

57/57 1.1s/it 1:03

Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7	1.5s/it 10.6s				
all	801	944	0.038	0.771	0.138	

0.116

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
2/30	7.22G	0.7401	2.649	1.093	136	640: 100%

57/57 1.1it/s 50.7s

Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7	1.5s/it 10.7s				
all	801	944	0.282	0.5	0.276	

0.218

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
3/30	7.23G	0.7407	2.265	1.083	140	640: 100%

57/57 1.1it/s 51.0s

Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7	1.5s/it 10.5s				
all	801	944	0.431	0.534	0.46	

0.368

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
4/30	7.25G	0.72	1.897	1.07	143	640: 100%

57/57 1.1it/s 51.4s

Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7	1.4s/it 9.7s				
all	801	944	0.62	0.615	0.65	

0.535

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
5/30	7.27G	0.7041	1.577	1.05	122	640: 100%

57/57 1.1it/s 50.1s

Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7	1.5s/it 10.7s				
all	801	944	0.722	0.643	0.742	

0.616

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
6/30	7.29G	0.6739	1.369	1.033	124	640: 100%

57/57 1.1it/s 51.5s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.4s/it	9.8s						
	all	801	944	0.765	0.661	0.761			
0.632									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
7/30	7.3G	0.6712	1.241	1.025	120	640: 100%			
57/57 1.1it/s 52.0s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.3s/it	9.4s						
	all	801	944	0.836	0.743	0.832			
0.698									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
8/30	7.32G	0.6383	1.12	1.015	138	640: 100%			
57/57 1.1it/s 52.9s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.4s/it	9.6s						
	all	801	944	0.864	0.76	0.855			
0.716									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
9/30	7.34G	0.6362	1.061	1.011	131	640: 100%			
57/57 1.1it/s 52.6s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.4s/it	9.5s						
	all	801	944	0.914	0.782	0.886			
0.744									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
10/30	7.35G	0.6221	0.981	1	160	640: 100%			
57/57 1.1it/s 52.4s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.3s/it	8.8s						
	all	801	944	0.878	0.813	0.898			
0.756									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
11/30	7.37G	0.6207	0.9413	0.9929	128	640: 100%			
57/57 1.1it/s 51.6s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.3s/it	9.0s						
	all	801	944	0.877	0.798	0.89			
0.757									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
12/30	7.21G	0.6047	0.8863	0.9969	122	640: 100%			
57/57 1.1it/s 51.3s									
	Class	Images	Instances	Box(P	R	mAP50	mAP50		
-95): 100%	7/7	1.4s/it	9.9s						
	all	801	944	0.921	0.803	0.897			
0.758									
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size			
13/30	7.21G	0.599	0.852	0.9913	136	640: 100%			

57/57 1.1it/s 51.6s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.5s/it 10.2s						
	all	801	944	0.913	0.815	0.9	
0.756							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
14/30	7.21G	0.5888	0.8157	0.9839	118	640: 100%	
57/57 1.1it/s 51.4s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.4s/it 10.1s						
	all	801	944	0.931	0.826	0.908	
0.772							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
15/30	7.21G	0.5852	0.7845	0.9784	135	640: 100%	
57/57 1.1it/s 51.9s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.2s/it 8.7s						
	all	801	944	0.919	0.843	0.912	
0.776							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
16/30	7.23G	0.57	0.7534	0.9699	130	640: 100%	
57/57 1.1it/s 50.5s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.4s/it 9.5s						
	all	801	944	0.935	0.85	0.922	
0.784							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
17/30	7.25G	0.5669	0.751	0.9702	150	640: 100%	
57/57 1.1it/s 51.9s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.4s/it 9.6s						
	all	801	944	0.925	0.846	0.932	
0.794							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
18/30	7.27G	0.5677	0.7316	0.9733	147	640: 100%	
57/57 1.1it/s 51.9s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.4s/it 10.1s						
	all	801	944	0.944	0.866	0.941	
0.8							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
19/30	7.28G	0.5578	0.7076	0.9632	124	640: 100%	
57/57 1.1it/s 51.6s							
	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	7/7 1.3s/it 9.1s						
	all	801	944	0.909	0.875	0.933	
0.798							
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
20/30	7.3G	0.5438	0.6811	0.9686	135	640: 100%	

```

57/57 1.1it/s 50.9s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.3s/it 9.2s
all        801      944      0.931      0.867      0.934

```

0.794

Closing dataloader mosaic

alumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
21/30       7.32G    0.5551    0.5124    0.9422      70      640: 100%
57/57 1.0it/s 56.3s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.3s/it 9.3s
all        801      944      0.922      0.881      0.942

```

0.797

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
22/30       7.34G    0.5478    0.4908    0.932      75      640: 100%
57/57 1.2it/s 48.7s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.3s/it 9.4s
all        801      944      0.934      0.878      0.942

```

0.804

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
23/30       7.35G    0.5399    0.458     0.9314      67      640: 100%
57/57 1.2it/s 48.7s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.3s/it 9.1s
all        801      944      0.923      0.904      0.95

```

0.803

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
24/30       7.37G    0.5349    0.4373    0.9242      73      640: 100%
57/57 1.1it/s 49.7s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.3s/it 9.1s
all        801      944      0.937      0.887      0.95

```

0.813

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
25/30       7.59G    0.5284    0.4199    0.9254      70      640: 100%
57/57 1.2it/s 48.8s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.4s/it 9.6s
all        801      944      0.923      0.906      0.956

```

0.82

```

Epoch      GPU_mem  box_loss  cls_loss  dfl_loss  Instances  Size
26/30       7.24G    0.5168    0.4075    0.9178      71      640: 100%
57/57 1.1it/s 50.8s
Class      Images  Instances  Box(P          R          mAP50  mAP50
-95): 100% 7/7 1.4s/it 9.6s
all        801      944      0.919      0.904      0.951

```

0.819

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
27/30	7.24G	0.5041	0.3932	0.9119	74	640: 100%
	57/57 1.1it/s	50.0s				
	Class	Images	Instances	Box(P	R	mAP50 mAP50
-95): 100%	7/7 1.5s/it		10.3s			
	all	801	944	0.964	0.884	0.956

0.827

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
28/30	7.24G	0.5039	0.3859	0.9091	75	640: 100%
	57/57 1.1it/s	49.6s				
	Class	Images	Instances	Box(P	R	mAP50 mAP50
-95): 100%	7/7 1.4s/it		9.8s			
	all	801	944	0.946	0.876	0.953

0.822

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
29/30	7.24G	0.4934	0.3754	0.9033	71	640: 100%
	57/57 1.1it/s	50.3s				
	Class	Images	Instances	Box(P	R	mAP50 mAP50
-95): 100%	7/7 1.2s/it		8.7s			
	all	801	944	0.956	0.89	0.957

0.827

Epoch	GPU_mem	box_loss	cls_loss	df1_loss	Instances	Size
30/30	7.24G	0.4884	0.3635	0.9014	76	640: 100%
	57/57 1.2it/s	49.1s				
	Class	Images	Instances	Box(P	R	mAP50 mAP50
-95): 100%	7/7 1.4s/it		9.5s			
	all	801	944	0.956	0.887	0.96

0.828

30 epochs completed in 0.519 hours.

Optimizer stripped from /content/runs/detect/train/weights/last.pt, 6.3MB

Optimizer stripped from /content/runs/detect/train/weights/best.pt, 6.3MB

Validating /content/runs/detect/train/weights/best.pt...

Ultralytics 8.3.248 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)

Model summary (fused): 72 layers, 3,008,573 parameters, 0 gradients, 8.1 GFLOPs

Class	Images	Instances	Box(P	R	mAP50 mAP50	
-95): 100%	7/7 2.1s/it		14.9s			
	all	801	944	0.955	0.887	0.96

0.828

Green Light	87	122	0.905	0.704	0.836
-------------	----	-----	-------	-------	-------

0.506

Red Light	74	108	0.912	0.671	0.831
-----------	----	-----	-------	-------	-------

0.499

Speed Limit 100	52	52	0.957	0.942	0.987
-----------------	----	----	-------	-------	-------

0.874

Speed Limit 110	17	17	0.761	0.882	0.952
-----------------	----	----	-------	-------	-------

0.872

Speed Limit 120	60	60	1	0.92	0.994
-----------------	----	----	---	------	-------

0.92

Speed Limit 20	56	56	0.987	0.982	0.986
----------------	----	----	-------	-------	-------

0.862	Speed Limit 30	71	74	0.993	0.959	0.987
0.931	Speed Limit 40	53	55	0.952	0.945	0.989
0.882	Speed Limit 50	68	71	1	0.847	0.975
0.874	Speed Limit 60	76	76	0.987	0.908	0.966
0.887	Speed Limit 70	78	78	0.978	0.962	0.988
0.893	Speed Limit 80	56	56	1	0.871	0.986
0.873	Speed Limit 90	38	38	0.969	0.842	0.971
0.804	Stop	81	81	0.974	0.988	0.992

0.915

Speed: 0.2ms preprocess, 2.4ms inference, 0.0ms loss, 7.3ms postprocess per image

Results saved to /content/runs/detect/train

- Validation step

```
In [11]: import os
import cv2
import matplotlib.pyplot as plt

def display_images(post_training_files_path, image_files):

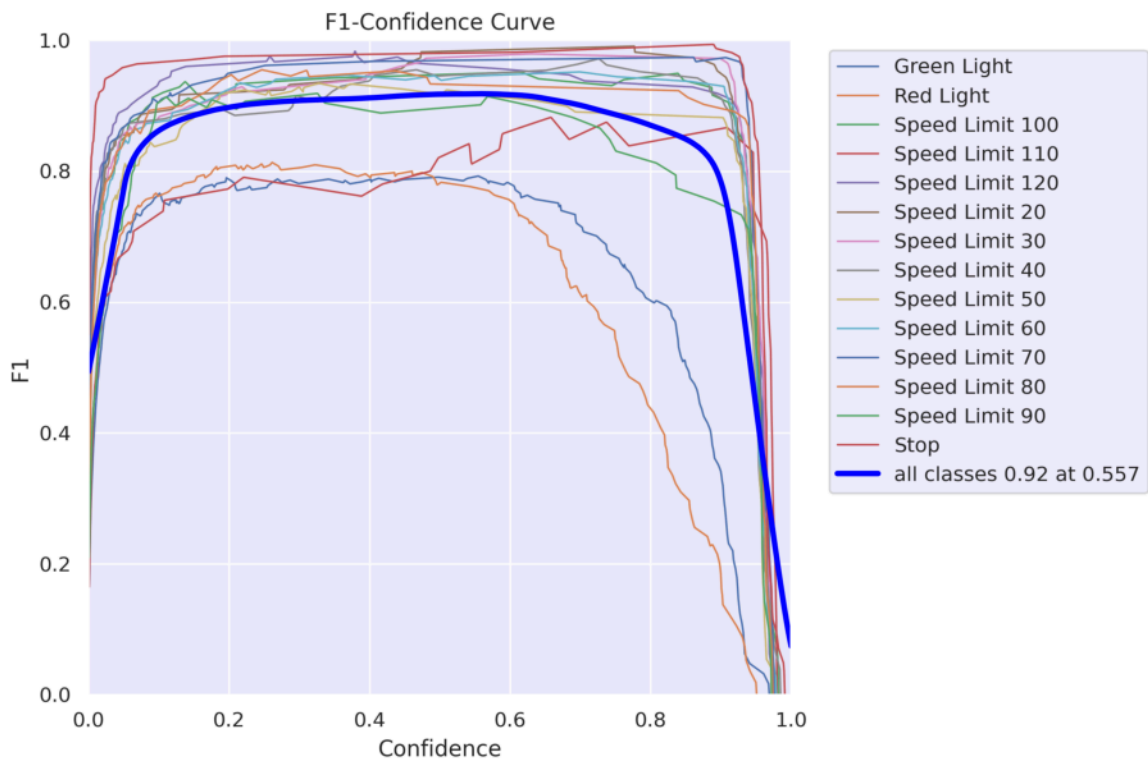
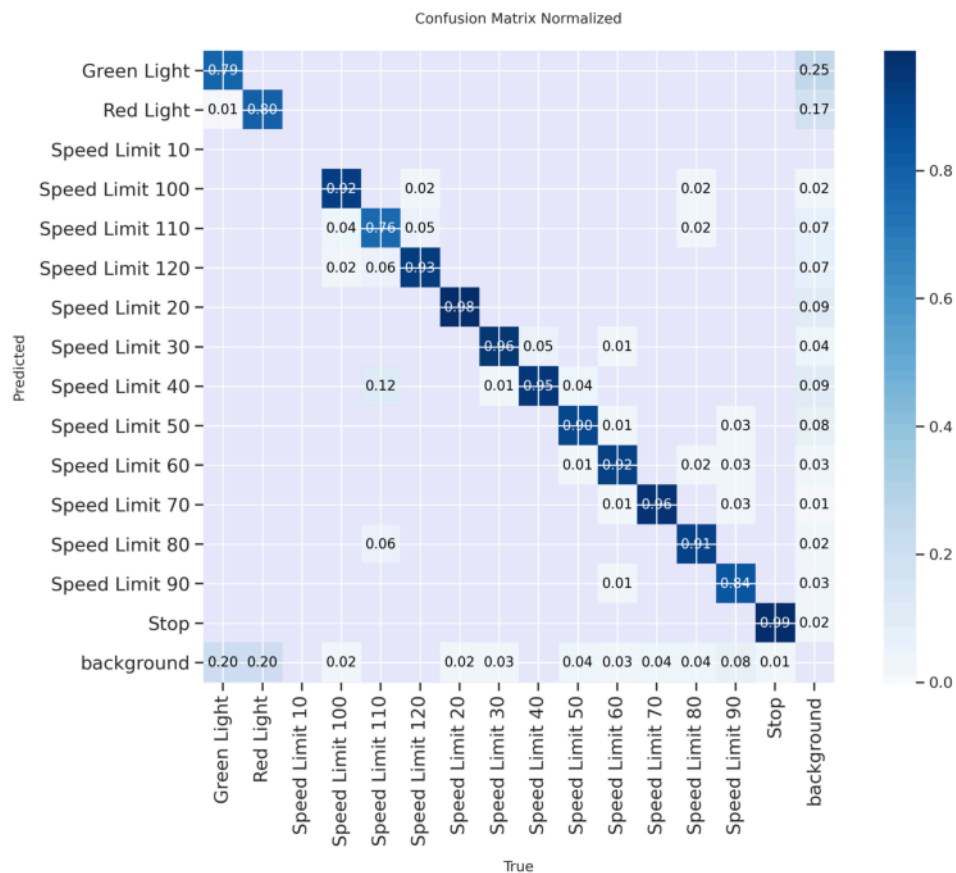
    for image_file in image_files:
        image_path = os.path.join(post_training_files_path, image_file)
        img = cv2.imread(image_path)
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

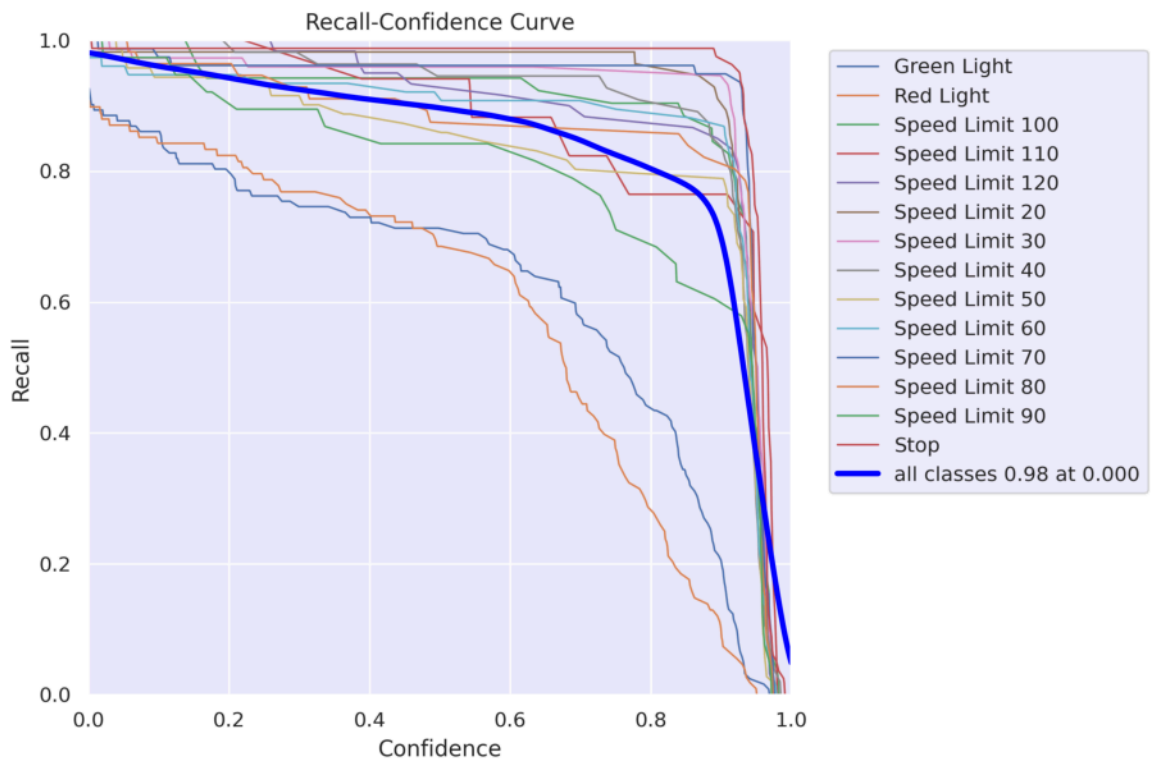
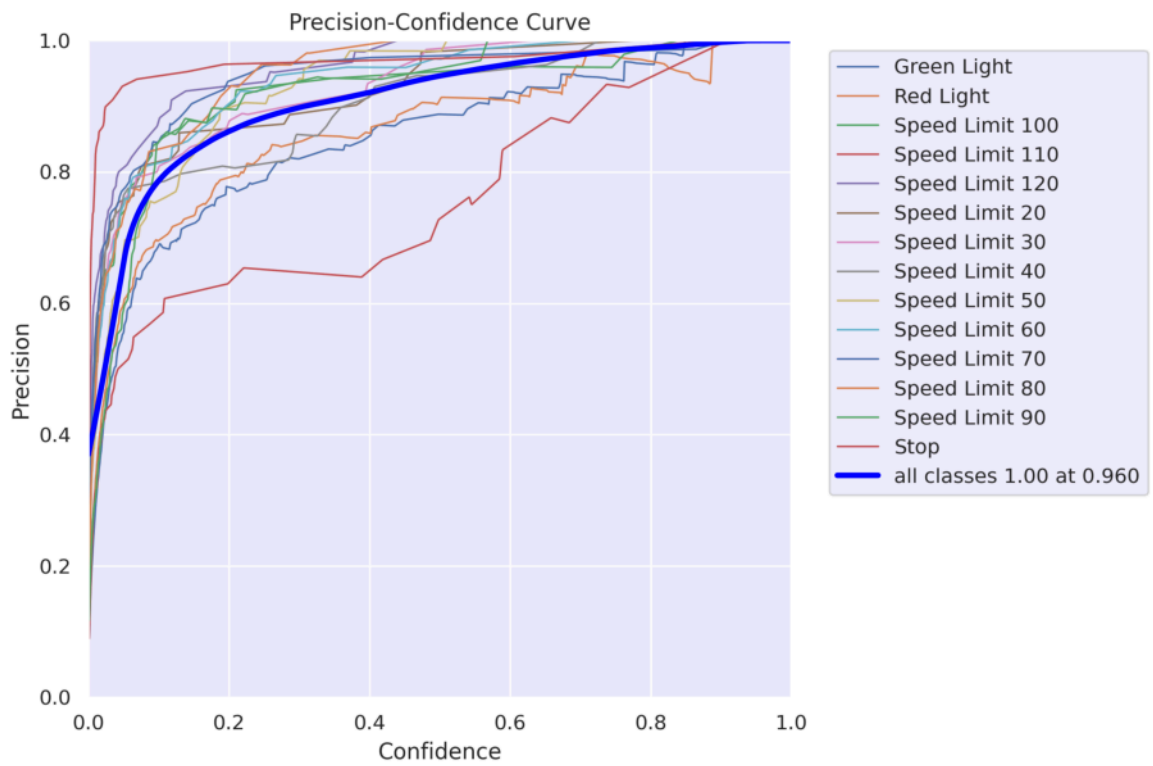
        plt.figure(figsize=(10, 10), dpi=120)
        plt.imshow(img)
        plt.axis('off')
        plt.show()

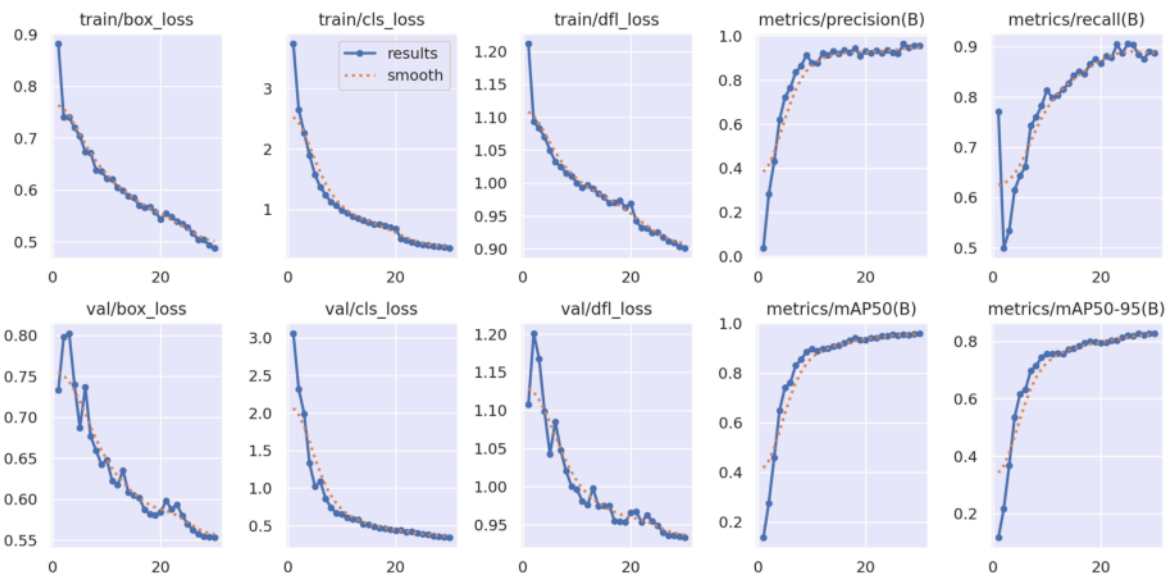
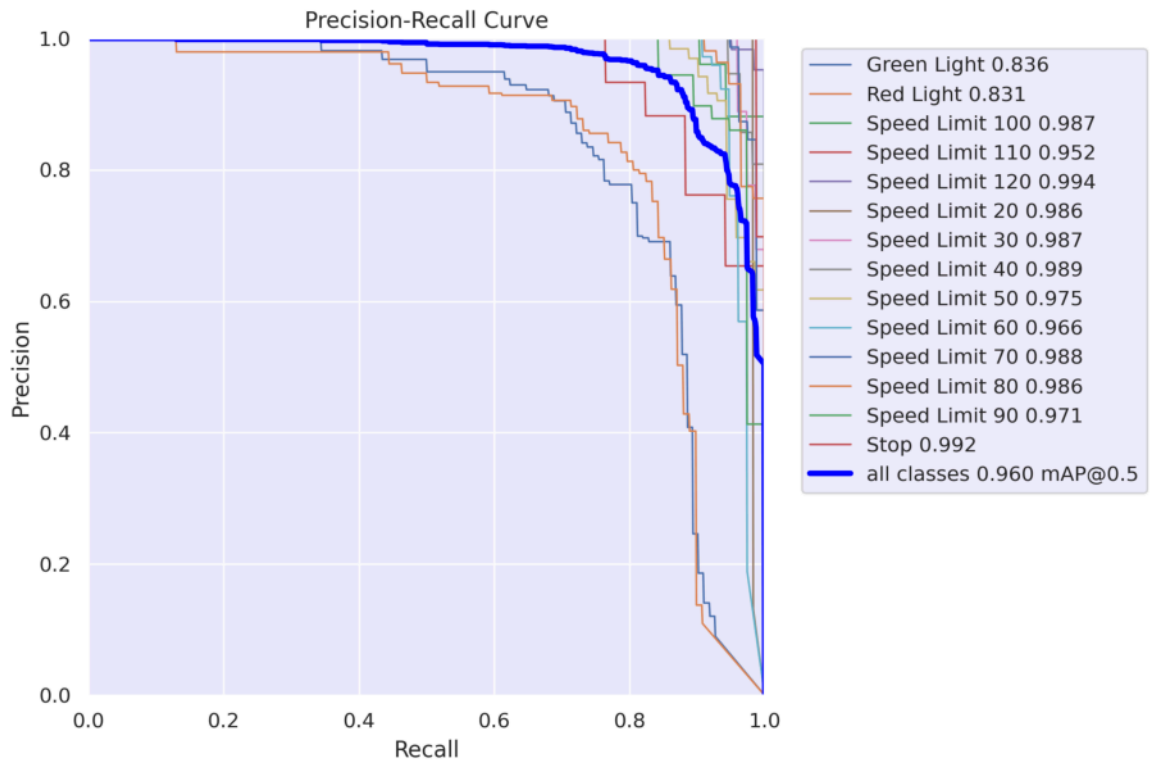
    # List of image files to display
    image_files = [
        'confusion_matrix_normalized.png',
        'BoxF1_curve.png',
        'BoxP_curve.png',
        'BoxR_curve.png',
        'BoxPR_curve.png',
        'results.png'
    ]

    # Path to the directory containing the images
    post_training_files_path = 'runs/detect/train'

    # Display the images
    display_images(post_training_files_path, image_files)
```







```
In [12]: Result_Final_model = pd.read_csv('runs/detect/train/results.csv')
Result_Final_model.tail(10)
```

Out[12]:

	epoch	time	train/box_loss	train/cls_loss	train/dfl_loss	metrics/precision(B)	metrics/recall(B)
20	21	1325.89	0.55514	0.51236	0.94222	0.92231	0.92231
21	22	1385.07	0.54780	0.49077	0.93199	0.93354	0.93354
22	23	1443.63	0.53989	0.45799	0.93142	0.92321	0.92321
23	24	1503.59	0.53490	0.43730	0.92417	0.93665	0.93665
24	25	1564.96	0.52842	0.41991	0.92537	0.92347	0.92347
25	26	1626.78	0.51676	0.40749	0.91780	0.91909	0.91909
26	27	1687.99	0.50409	0.39320	0.91191	0.96447	0.96447
27	28	1748.23	0.50394	0.38592	0.90912	0.94588	0.94588
28	29	1808.03	0.49342	0.37543	0.90331	0.95584	0.95584
29	30	1867.23	0.48843	0.36354	0.90136	0.95584	0.95584



In [13]:

```
# Read the results.csv file as a pandas dataframe
Result_Final_model.columns = Result_Final_model.columns.str.strip()

# Create subplots
fig, axs = plt.subplots(nrows=5, ncols=2, figsize=(15, 15))

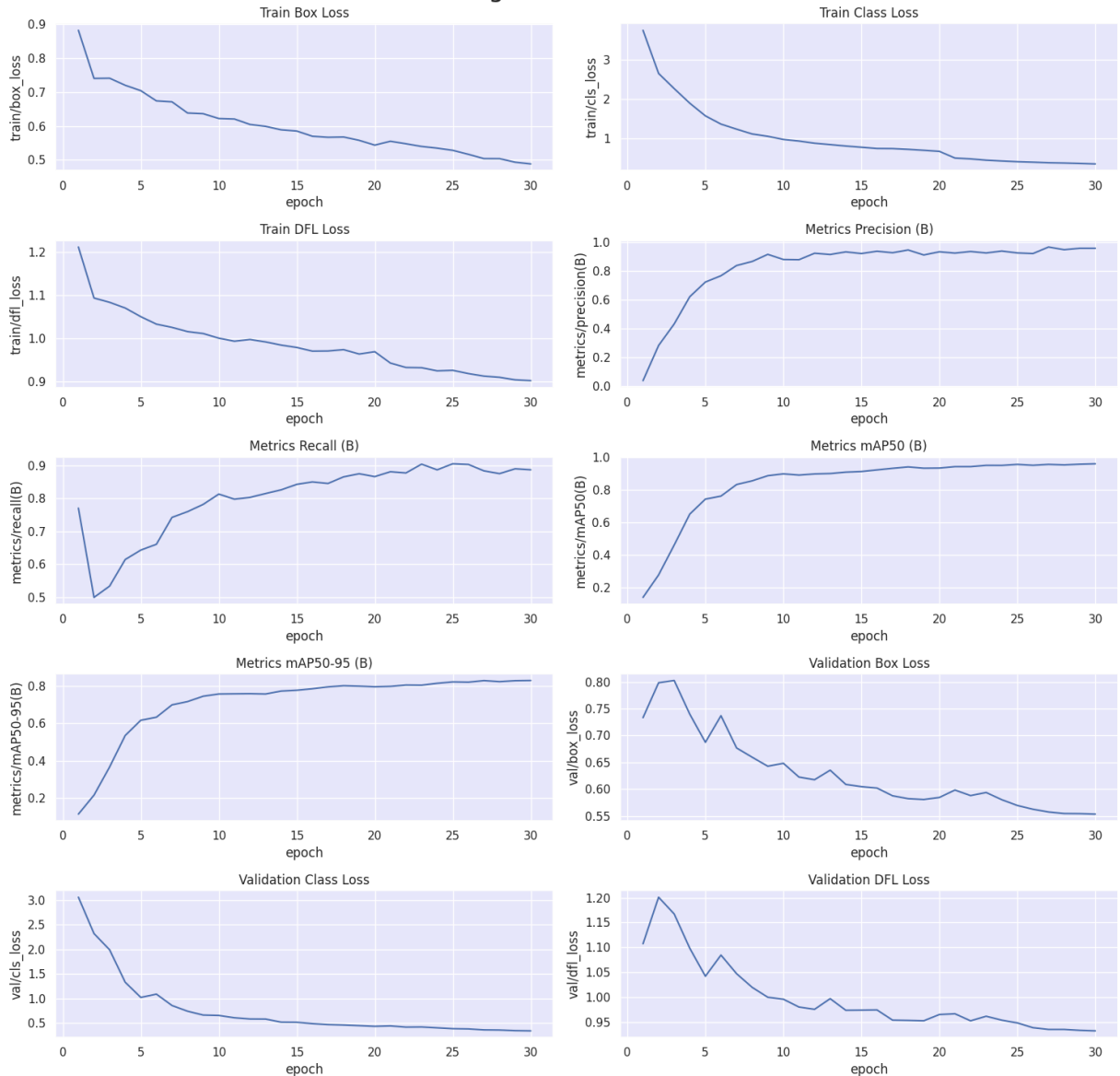
# Plot the columns using seaborn
sns.lineplot(x='epoch', y='train/box_loss', data=Result_Final_model, ax=axs[0,0])
sns.lineplot(x='epoch', y='train/cls_loss', data=Result_Final_model, ax=axs[0,1])
sns.lineplot(x='epoch', y='train/dfl_loss', data=Result_Final_model, ax=axs[1,0])
sns.lineplot(x='epoch', y='metrics/precision(B)', data=Result_Final_model, ax=axs[1,1])
sns.lineplot(x='epoch', y='metrics/recall(B)', data=Result_Final_model, ax=axs[2,0])
sns.lineplot(x='epoch', y='metrics/mAP50(B)', data=Result_Final_model, ax=axs[2,1])
sns.lineplot(x='epoch', y='metrics/mAP50-95(B)', data=Result_Final_model, ax=axs[3,0])
sns.lineplot(x='epoch', y='val/box_loss', data=Result_Final_model, ax=axs[3,1])
sns.lineplot(x='epoch', y='val/cls_loss', data=Result_Final_model, ax=axs[4,0])
sns.lineplot(x='epoch', y='val/dfl_loss', data=Result_Final_model, ax=axs[4,1])

# Set titles and axis labels for each subplot
axs[0,0].set(title='Train Box Loss')
axs[0,1].set(title='Train Class Loss')
axs[1,0].set(title='Train DFL Loss')
axs[1,1].set(title='Metrics Precision (B)')
axs[2,0].set(title='Metrics Recall (B)')
axs[2,1].set(title='Metrics mAP50 (B)')
axs[3,0].set(title='Metrics mAP50-95 (B)')
axs[3,1].set(title='Validation Box Loss')
axs[4,0].set(title='Validation Class Loss')
axs[4,1].set(title='Validation DFL Loss')

plt.suptitle('Training Metrics and Loss', fontsize=24)
plt.subplots_adjust(top=0.8)
```

```
plt.tight_layout()
plt.show()
```

Training Metrics and Loss



```
In [14]: # Loading the best performing model
Valid_model = YOLO('runs/detect/train/weights/best.pt')

# Evaluating the model on the validset
metrics = Valid_model.val(split = 'val')

# final results
print("precision(B): ", metrics.results_dict["metrics/precision(B)"])
print("metrics/recall(B): ", metrics.results_dict["metrics/recall(B)"])
print("metrics/mAP50(B): ", metrics.results_dict["metrics/mAP50(B)"])
print("metrics/mAP50-95(B): ", metrics.results_dict["metrics/mAP50-95(B)"])
```

Ultralytics 8.3.248 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
 Model summary (fused): 72 layers, 3,008,573 parameters, 0 gradients, 8.1 GFLOPs
 val: Fast image access ✅ (ping: 0.0±0.0 ms, read: 395.1±217.7 MB/s, size: 11.2 KB)
 val: Scanning /content/car/valid/labels.cache... 801 images, 0 backgrounds, 0 corrupt
 t: 100% ————— 801/801 970.7Kit/s 0.0s

	Class	Images	Instances	Box(P	R	mAP50	mAP50
-95): 100%	all	51/51	6.0it/s	8.4s			
0.828		801	944	0.956	0.887	0.96	
0.503	Green Light	87	122	0.905	0.703	0.832	
0.5	Red Light	74	108	0.912	0.671	0.832	
0.88	Speed Limit 100	52	52	0.957	0.942	0.987	
0.865	Speed Limit 110	17	17	0.763	0.882	0.952	
0.914	Speed Limit 120	60	60	1	0.92	0.994	
0.858	Speed Limit 20	56	56	0.987	0.982	0.986	
0.931	Speed Limit 30	71	74	0.993	0.959	0.987	
0.889	Speed Limit 40	53	55	0.952	0.945	0.989	
0.872	Speed Limit 50	68	71	1	0.847	0.975	
0.884	Speed Limit 60	76	76	0.988	0.908	0.966	
0.897	Speed Limit 70	78	78	0.978	0.962	0.988	
0.878	Speed Limit 80	56	56	1	0.871	0.986	
0.804	Speed Limit 90	38	38	0.972	0.842	0.971	
0.918	Stop	81	81	0.974	0.988	0.992	

Speed: 1.5ms preprocess, 4.1ms inference, 0.0ms loss, 1.3ms postprocess per image
 Results saved to /content/runs/detect/val
 precision(B): 0.9558580165453353
 metrics/recall(B): 0.8873978668025
 metrics/mAP50(B): 0.9597801490176275
 metrics/mAP50-95(B): 0.828185550240818

- Make predictions on test images

```
In [15]: # Normalization function
def normalize_image(image):
    return image / 255.0

# Image resizing function
def resize_image(image, size=(640, 640)):
    return cv2.resize(image, size)

# Path to validation images
```

```

dataset_path = 'car' # Place your dataset path here
valid_images_path = os.path.join(dataset_path, 'test', 'images')

# List of all jpg images in the directory
image_files = [file for file in os.listdir(valid_images_path) if file.endswith('.jpg')]

# Check if there are images in the directory
if len(image_files) > 0:
    # Select 9 images at equal intervals
    num_images = len(image_files)
    step_size = max(1, num_images // 9) # Ensure the interval is at least 1
    selected_images = [image_files[i] for i in range(0, num_images, step_size)]

    # Prepare subplots
    fig, axes = plt.subplots(3, 3, figsize=(20, 21))
    fig.suptitle('Validation Set Inferences', fontsize=24)

    for i, ax in enumerate(axes.flatten()):
        if i < len(selected_images):
            image_path = os.path.join(valid_images_path, selected_images[i])

            # Load image
            image = cv2.imread(image_path)

            # Check if the image is loaded correctly
            if image is not None:
                # Resize image
                resized_image = resize_image(image, size=(640, 640))
                # Normalize image
                normalized_image = normalize_image(resized_image)

                # Convert the normalized image to uint8 data type
                normalized_image_uint8 = (normalized_image * 255).astype(np.uint8)

                # Predict with the model
                results = Valid_model.predict(source=normalized_image_uint8, imgsiz=

                # Plot image with labels
                annotated_image = results[0].plot(line_width=1)
                annotated_image_rgb = cv2.cvtColor(annotated_image, cv2.COLOR_BGR2RGB)
                ax.imshow(annotated_image_rgb)
            else:
                print(f"Failed to load image {image_path}")
            ax.axis('off')

    plt.tight_layout()
    plt.show()

```


0: 640x640 1 Red Light, 11.2ms

Speed: 2.2ms preprocess, 11.2ms inference, 9.7ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 40, 1 Stop, 7.5ms

Speed: 2.4ms preprocess, 7.5ms inference, 1.4ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 30, 7.5ms

Speed: 2.5ms preprocess, 7.5ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 30, 7.8ms

Speed: 3.6ms preprocess, 7.8ms inference, 1.8ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 100, 7.4ms

Speed: 2.7ms preprocess, 7.4ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 50, 7.5ms

Speed: 2.6ms preprocess, 7.5ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 120, 7.5ms

Speed: 2.8ms preprocess, 7.5ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 1 Speed Limit 120, 7.4ms

Speed: 3.3ms preprocess, 7.4ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

0: 640x640 2 Speed Limit 40s, 1 Speed Limit 60, 7.4ms

Speed: 3.8ms preprocess, 7.4ms inference, 1.3ms postprocess per image at shape (1, 3, 640, 640)

Validation Set Inferences



Documentation

Summary

In this project we are trying to classify traffic signs from images.

To do so, we will be using a YOLO model in order to make predictions on the images so they can be classified into different types of traffic signs.

Implementation

First step is fetching the data and exploring it. As shown, we are using images of different resolutions that contain traffic signs such as speed limits, traffic lights and stops.

For a baseline we use the pre-trained YOLOv8 model to predict a speed limit, but as shown it does not detect it.

The next step is training the actual model. This will be done in 30 epochs, each with 50 steps and an auto optimizer.

After the 30 epochs we can see a precision of 95.58%, recall of 88.74%, mAP50 of 95.97% and mAP50-95 of 82.81%.

Results

When using the model on the test images, we can see that it almost predicts all inputs correctly, with the exception of a 20 speed limit which it sees as a 120 speed limit.